

AP FRQ Pack

1. Fish enter a lake at a rate modeled by the function E given by $E(t) = 20 + 15 \sin\left(\frac{\pi t}{6}\right)$. Fish leave the lake at a rate modeled by the function L given by $L(t) = 4 + 2^{0.1t^2}$. Both $E(t)$ and $L(t)$ are measured in fish per hour, and t is measured in hours since midnight ($t = 0$).
 - (a) How many fish enter the lake over the 5-hour period from midnight ($t = 0$) to 5 A.M. ($t = 5$)? Give your answer to the nearest whole number.
 - (b) What is the average number of fish that leave the lake per hour over the 5-hour period from midnight ($t = 0$) to 5 A.M. ($t = 5$)?
 - (c) At what time t , for $0 \leq t \leq 8$, is the greatest number of fish in the lake? Justify your answer.
 - (d) Is the rate of change in the number of fish in the lake increasing or decreasing at 5 A.M. ($t = 5$)? Explain your reasoning.

Solution: Part (a): $\int_0^5 E(t) dt = 153.457690$

To the nearest whole number, 153 fish enter the lake from midnight to 5 A.M.

Part (b): $\frac{1}{5-0} \int_0^5 L(t) dt = 6.059038$

The average number of fish that leave the lake per hour from midnight to 5 A.M. is 6.059 fish per hour.

Part (c): The rate of change in the number of fish in the lake at time t is given by $E(t) - L(t)$.

$$E(t) - L(t) = 0 \Rightarrow t = 6.20356$$

$E(t) - L(t) > 0$ for $0 \leq t < 6.20356$, and $E(t) - L(t) < 0$ for $6.20356 < t \leq 8$. Therefore the greatest number of fish in the lake is at time $t = 6.204$ (or 6.203).

— OR —

Let $A(t)$ be the change in the number of fish in the lake from midnight to t hours after midnight.

$$A(t) = \int_0^t (E(s) - L(s)) ds$$

$$A'(t) = E(t) - L(t) = 0 \Rightarrow t = C = 6.20356$$

Therefore the greatest number of fish in the lake is at time $t = 6.204$ (or 6.203).

Part (d): $E'(5) - L'(5) = -10.7228 < 0$

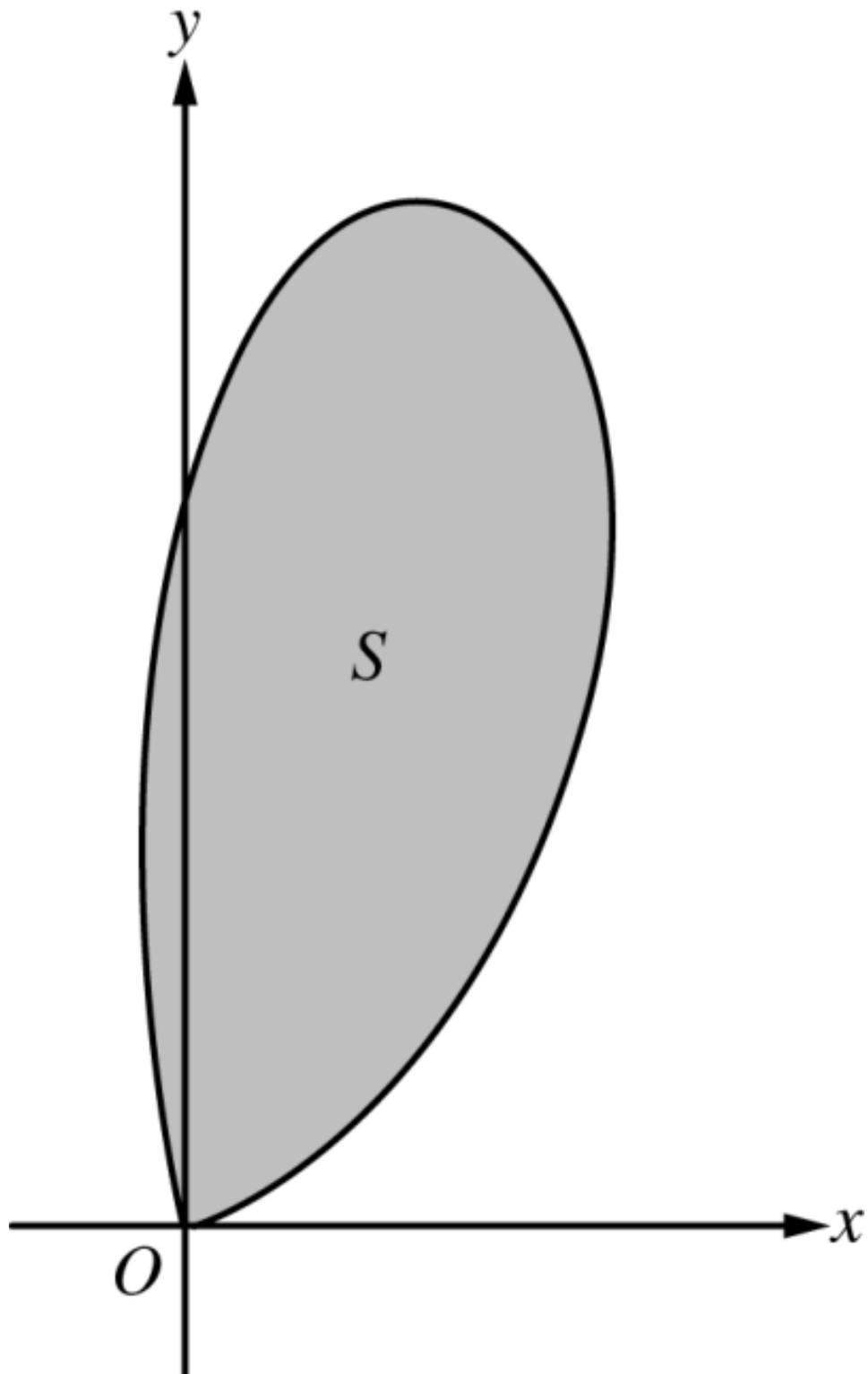
Because $E'(5) - L'(5) < 0$, the rate of change in the number of fish is decreasing at time $t = 5$.

t	$A(t)$
0	0
C	135.01492
8	80.91998

Scoring Guide

Part (a): 2 : { 1 : integral, 1 : answer } Part (b): 2 : { 1 : integral, 1 : answer } Part (c): 3 : { 1 : sets $E(t) - L(t) = 0$, 1 : answer, 1 : justification } Part (d): 2 : { 1 : considers $E'(5)$ and $L'(5)$, 1 : answer with explanation }

2. Let S be the region bounded by the graph of the polar curve $r(\theta) = 3\sqrt{\theta} \sin(\theta^2)$ for $0 \leq \theta \leq \sqrt{\pi}$, as shown in the figure above.



- (a) Find the area of S .
- (b) What is the average distance from the origin to a point on the polar curve $r(\theta) = 3\sqrt{\theta} \sin(\theta^2)$ for $0 \leq \theta \leq \sqrt{\pi}$?

- (c) There is a line through the origin with positive slope m that divides the region S into two regions with equal areas. Write, but do not solve, an equation involving one or more integrals whose solution gives the value of m .
- (d) For $k > 0$, let $A(k)$ be the area of the portion of region S that is also inside the circle $r = k \cos \theta$. Find $\lim_{k \rightarrow \infty} A(k)$.

Solution: Part (a): $\frac{1}{2} \int_0^{\sqrt{\pi}} (r(\theta))^2 d\theta = 3.534292$

The area of S is 3.534.

Part (b): $\frac{1}{\sqrt{\pi}-0} \int_0^{\sqrt{\pi}} r(\theta) d\theta = 1.579933$

The average distance from the origin to a point on the curve $r = r(\theta)$ for $0 \leq \theta \leq \sqrt{\pi}$ is 1.580 (or 1.579).

Part (c): $\tan \theta = \frac{y}{x} = m \Rightarrow \theta = \tan^{-1} m$

$$\frac{1}{2} \int_0^{\tan^{-1} m} (r(\theta))^2 d\theta = \frac{1}{2} \left(\frac{1}{2} \int_0^{\sqrt{\pi}} (r(\theta))^2 d\theta \right)$$

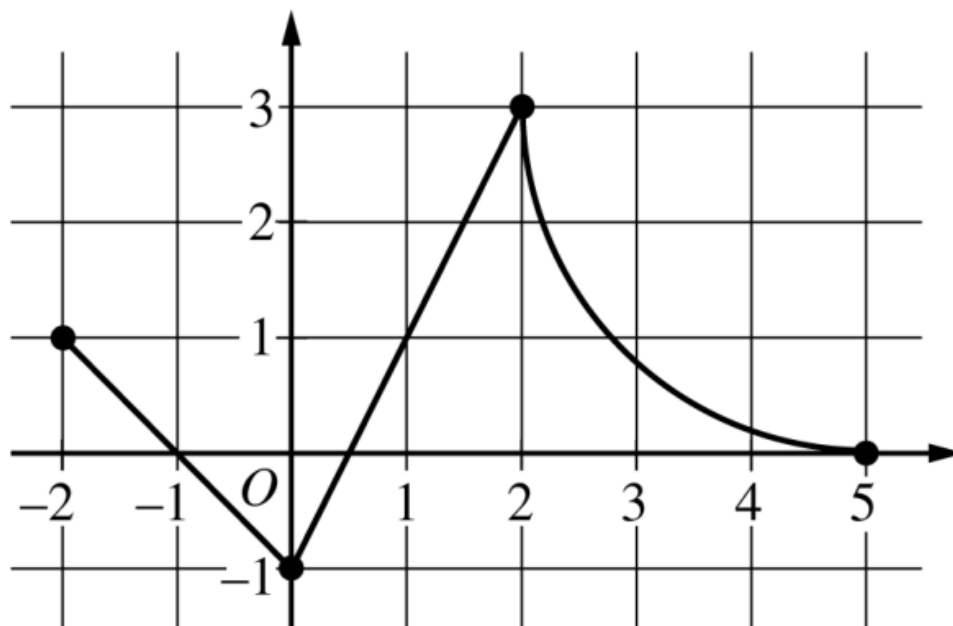
Part (d): As $k \rightarrow \infty$, the circle $r = k \cos \theta$ grows to enclose all points to the right of the y -axis.

$$\lim_{k \rightarrow \infty} A(k) = \frac{1}{2} \int_0^{\pi/2} (r(\theta))^2 d\theta = \frac{1}{2} \int_0^{\pi/2} \left(3\sqrt{\theta} \sin(\theta^2) \right)^2 d\theta = 3.324$$

Scoring Guide

Part (a): 2 points: {1 point: integral; 1 point: answer} Part (b): 2 points: {1 point: integral; 1 point: answer} Part (c): 3 points: {1 point: equates polar areas; 1 point: inverse trigonometric function applied to m as limit of integration; 1 point: equation} Part (d): 2 points: {1 point: limits of integration; 1 point: answer with integral}

3. The continuous function f is defined on the closed interval $-6 \leq x \leq 5$. The figure above shows a portion of the graph of f , consisting of two line segments and a quarter of a circle centered at the point $(5, 3)$. It is known that the point $(3, 3 - \sqrt{5})$ is on the graph of f .



Graph of f

Graph of f

- (a) If $\int_{-6}^5 f(x) dx = 7$, find the value of $\int_{-6}^{-2} f(x) dx$. Show the work that leads to your answer.
- (b) Evaluate $\int_3^5 (2f'(x) + 4) dx$.
- (c) The function g is given by $g(x) = \int_{-2}^x f(t) dt$. Find the absolute maximum value of g on the interval $-2 \leq x \leq 5$. Justify your answer.
- (d) Find $\lim_{x \rightarrow 1} \frac{10^x - 3f'(x)}{f(x) - \arctan x}$.

Solution: Part (a): $\int_{-6}^5 f(x) dx = \int_{-6}^{-2} f(x) dx + \int_{-2}^5 f(x) dx$

$$\Rightarrow 7 = \int_{-6}^{-2} f(x) dx + 2 + \left(9 - \frac{9\pi}{4}\right)$$

$$\Rightarrow \int_{-6}^{-2} f(x) dx = 7 - \left(11 - \frac{9\pi}{4}\right) = \frac{9\pi}{4} - 4$$

Part (b): $\int_3^5 (2f'(x) + 4) dx = 2 \int_3^5 f'(x) dx + \int_3^5 4 dx$

$$= 2(f(5) - f(3)) + 4(5 - 3)$$

$$= 2(0 - (3 - \sqrt{5})) + 8$$

$$= 2(-3 + \sqrt{5}) + 8 = 2 + 2\sqrt{5}$$

— OR —

$$\int_3^5 (2f'(x) + 4) dx = [2f(x) + 4x]_{x=3}^{x=5}$$

$$= (2f(5) + 20) - (2f(3) + 12)$$

$$= (2 \cdot 0 + 20) - (2(3 - \sqrt{5}) + 12)$$

$$= 2 + 2\sqrt{5}$$

Part (c): $g'(x) = f(x) = 0 \Rightarrow x = -1, x = \frac{1}{2}, x = 5$

Table of values:

On the interval $-2 \leq x \leq 5$, the absolute maximum value of g is $g(5) = 11 - \frac{9\pi}{4}$.

$$\begin{aligned} \text{Part (d): } \lim_{x \rightarrow 1} \frac{10^3 - 3f'(x)}{f(x) - \arctan x} &= \frac{10^3 - 3f'(1)}{f(1) - \arctan 1} \\ &= \frac{10 - 3 \cdot 2}{1 - \arctan 1} = \frac{4}{1 - \frac{\pi}{4}} \end{aligned}$$

x	$g(x)$
-2	0
-1	$\frac{1}{2}$
$\frac{1}{2}$	$-\frac{1}{4}$
5	$11 - \frac{9\pi}{4}$

Scoring Guide

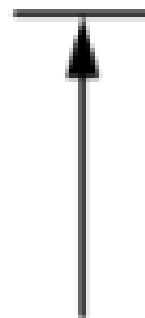
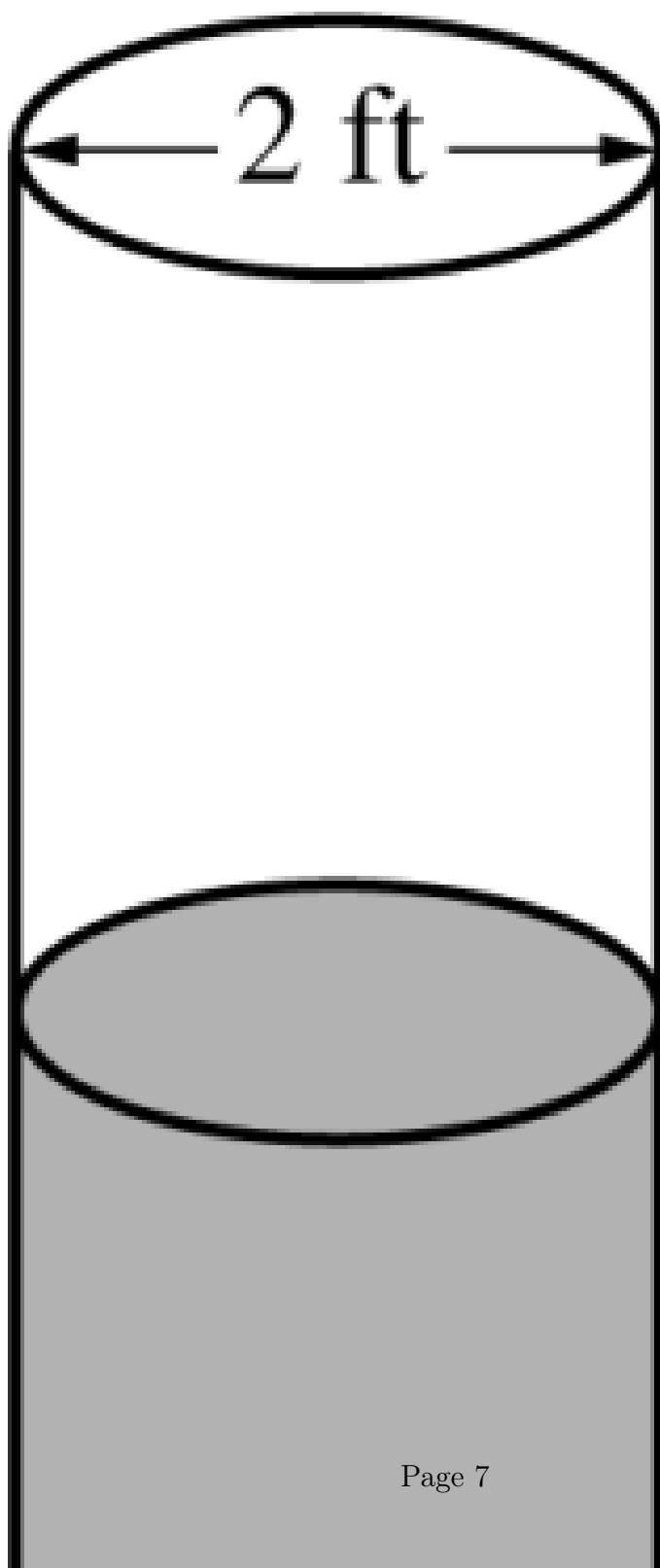
Part (a): 3 points 1 : $\int_{-6}^5 f(x) dx = \int_{-6}^{-2} f(x) dx + \int_{-2}^5 f(x) dx$ 1 : $\int_{-2}^5 f(x) dx$ 1 : answer

Part (b): 2 points 1 : Fundamental Theorem of Calculus 1 : answer Part (c): 3 points

1 : $g'(x) = f(x)$ 1 : identifies $x = -1$ as a candidate 1 : answer with justification Part

(d): 1 point 1 : answer

4. A cylindrical barrel with a diameter of 2 feet contains collected rainwater, as shown in the figure above. The water drains out through a valve (not shown) at the bottom of the barrel. The rate of change of the height h of the water in the barrel with respect to time t is modeled by $\frac{dh}{dt} = -\frac{1}{10}\sqrt{h}$, where h is measured in feet and t is measured in seconds. (The volume V of a cylinder with radius r and height h is $V = \pi r^2 h$.)



h ft

Cylindrical barrel with water

- (a) Find the rate of change of the volume of water in the barrel with respect to time when the height of the water is 4 feet. Indicate units of measure.
- (b) When the height of the water is 3 feet, is the rate of change of the height of the water with respect to time increasing or decreasing? Explain your reasoning.
- (c) At time $t = 0$ seconds, the height of the water is 5 feet. Use separation of variables to find an expression for h in terms of t .

Solution: Part (a): $V = \pi r^2 h = \pi(1)^2 h = \pi h$

$$\frac{dV}{dt} = \pi \frac{dh}{dt} \Big|_{h=4} = \pi \left(-\frac{1}{10} \sqrt{4} \right) = -\frac{\pi}{5} \text{ cubic feet per second}$$

$$\text{Part (b): } \frac{d^2 h}{dt^2} = -\frac{1}{20\sqrt{h}} \frac{dh}{dt} = -\frac{1}{20\sqrt{h}} \cdot \left(-\frac{1}{10} \sqrt{h} \right) = \frac{1}{200}$$

Because $\frac{d^2 h}{dt^2} = \frac{1}{200} > 0$ for $h > 0$, the rate of change of the height is increasing when the height of the water is 3 feet.

$$\text{Part (c): } \frac{dh}{\sqrt{h}} = -\frac{1}{10} dt$$

$$\int \frac{dh}{\sqrt{h}} = \int -\frac{1}{10} dt$$

$$2\sqrt{h} = -\frac{1}{10}t + C$$

$$2\sqrt{5} = -\frac{1}{10}(0) + C \Rightarrow C = 2\sqrt{5}$$

$$2\sqrt{h} = -\frac{1}{10}t + 2\sqrt{5}$$

$$h(t) = \left(-\frac{1}{20}t + \sqrt{5} \right)^2$$

Scoring Guide

Part (a): 2 points: 1 : $\frac{dV}{dt} = \pi \frac{dh}{dt}$ 1 : answer with units Part (b): 3 points: 1 : $\frac{d}{dh} \left(\frac{1}{10} \sqrt{h} \right) = -\frac{1}{20\sqrt{h}}$ 1 : $\frac{d^2 h}{dt^2} = -\frac{1}{20\sqrt{h}} \cdot \frac{dh}{dt}$ 1 : answer with explanation Part (c): 4 points: 1 : separation of variables 1 : antiderivatives 1 : constant of integration and uses initial condition 1 : $h(t)$ Note: 0/4 if no separation of variables Note: max 2/4 [1-1-0-0] if no constant of integration

5. Consider the family of functions $f(x) = \frac{1}{x^2 - 2x + k}$, where k is a constant.
 - (a) Find the value of k , for $k > 0$, such that the slope of the line tangent to the graph of f at $x = 0$ equals 6.
 - (b) For $k = -8$, find the value of $\int_0^1 f(x) dx$.
 - (c) For $k = 1$, find the value of $\int_0^2 f(x) dx$ or show that it diverges.

$$\text{Solution: Part (a): } f'(0) = \frac{2}{k^2} = 6 \Rightarrow k^2 = \frac{1}{3} \Rightarrow k = \frac{1}{\sqrt{3}}$$

$$\text{Part (b): } \frac{1}{x^2 - 2x - 8} = \frac{1}{(x-4)(x+2)} = \frac{A}{x-4} + \frac{B}{x+2}$$

$$\Rightarrow 1 = A(x + 2) + B(x - 4)$$

$$\Rightarrow A = \frac{1}{6}, B = -\frac{1}{6}$$

$$\int_0^1 f(x) dx = \int_0^1 \left(\frac{1}{6(x-4)} - \frac{1}{6(x+2)} \right) dx$$

$$= \left[\frac{1}{6} \ln |x - 4| - \frac{1}{6} \ln |x + 2| \right]_{x=0}^{x=1}$$

$$= \left(\frac{1}{6} \ln 3 - \frac{1}{6} \ln 3 \right) - \left(\frac{1}{6} \ln 4 - \frac{1}{6} \ln 2 \right) = -\frac{1}{6} \ln 2$$

$$\text{Part (c): } \int_0^2 \frac{1}{x^2 - 2x + 1} dx = \int_0^2 \frac{1}{(x-1)^2} dx = \int_0^1 \frac{1}{(x-1)^2} dx + \int_1^2 \frac{1}{(x-1)^2} dx$$

$$= \lim_{b \rightarrow 1^-} \int_0^b \frac{1}{(x-1)^2} dx + \lim_{b \rightarrow 1^+} \int_b^2 \frac{1}{(x-1)^2} dx$$

$$= \lim_{b \rightarrow 1^-} \left[-\frac{1}{x-1} \right]_{x=0}^{x=b} + \lim_{b \rightarrow 1^+} \left[-\frac{1}{x-1} \right]_{x=b}^{x=2}$$

$$= \lim_{b \rightarrow 1^-} \left(-\frac{1}{b-1} - 1 \right) + \lim_{b \rightarrow 1^+} \left(-1 + \frac{1}{b-1} \right)$$

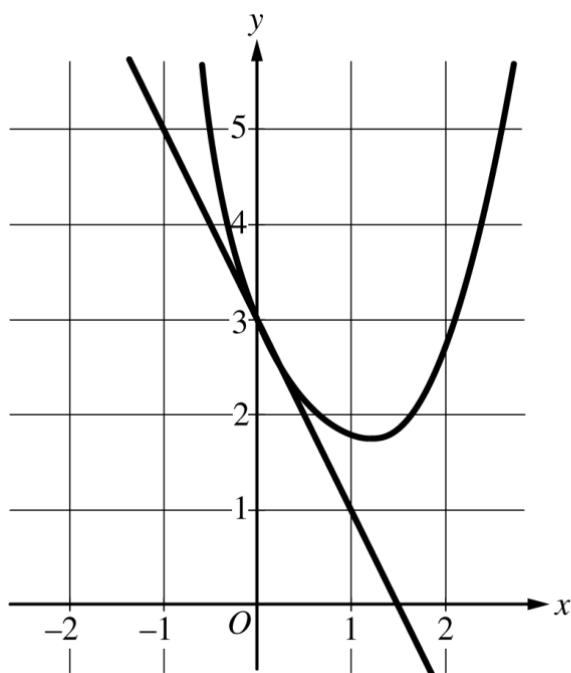
Because $\lim_{b \rightarrow 1^-} \left(-\frac{1}{b-1} \right)$ does not exist, the integral diverges.

Scoring Guide

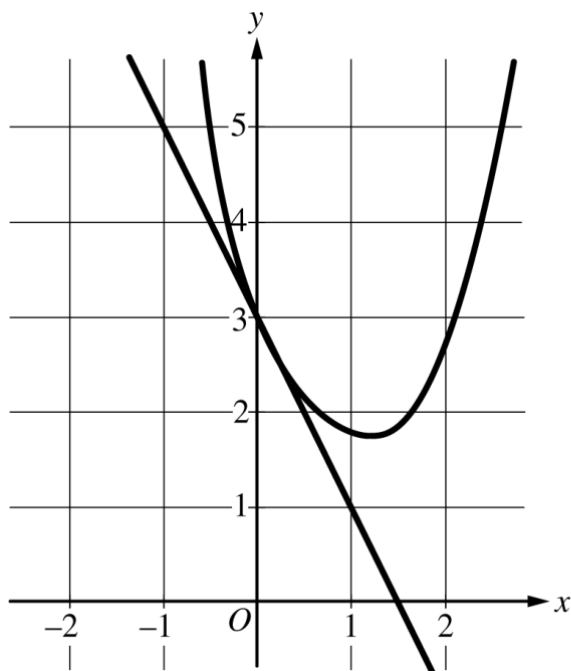
Part (a): 3 points { 1 : denominator of $f'(x)$ 1 : $f'(x)$ 1 : answer } Part (b): 3 points { 1 : partial fraction decomposition 1 : antiderivatives 1 : answer } Part (c): 3 points { 1 : improper integral 1 : antiderivative 1 : answer with reason }

6. A function f has derivatives of all orders for all real numbers x . A portion of the graph of f is shown above, along with the line tangent to the graph of f at $x = 0$. Selected derivatives of f at $x = 0$ are given in the table above.

n	$f^{(n)}(0)$
2	3
3	$-\frac{23}{2}$
4	54



n	$f^{(n)}(0)$
2	3
3	$-\frac{23}{2}$
4	54



n	$f^{(n)}(0)$
2	3
3	$-\frac{23}{2}$
4	54

- Write the third-degree Taylor polynomial for f about $x = 0$.
- Write the first three nonzero terms of the Maclaurin series for e^x . Write the second-degree Taylor polynomial for $e^x f(x)$ about $x = 0$.
- Let h be the function defined by $h(x) = \int_0^x f(t) dt$. Use the Taylor polynomial found in part (a) to find an approximation for $h(1)$.
- It is known that the Maclaurin series for h converges to $h(x)$ for all real numbers x . It is also known that the individual terms of the series for $h(1)$ alternate in sign

and decrease in absolute value to 0. Use the alternating series error bound to show that the approximation found in part (c) differs from $h(1)$ by at most 0.45.

Solution: Part (a): $f(0) = 3$ and $f'(0) = -2$. The third-degree Taylor polynomial for f about $x = 0$ is $3 - 2x + \frac{3}{2!}x^2 + \frac{-23}{3!}x^3 = 3 - 2x + \frac{3}{2}x^2 - \frac{23}{12}x^3$.

Part (b): The first three nonzero terms of the Maclaurin series for e^x are $1 + x + \frac{1}{2!}x^2$. The second-degree Taylor polynomial for $e^x f(x)$ about $x = 0$ is $3\left(1 + x + \frac{1}{2!}x^2\right) - 2x(1 + x) + \frac{3}{2}x^2(1) = 3 + (3 - 2)x + \left(\frac{3}{2} - 2 + \frac{3}{2}\right)x^2 = 3 + x + x^2$.

Part (c): $h(1) = \int_0^1 f(t) dt \approx \int_0^1 \left(3 - 2t + \frac{3}{2}t^2 - \frac{23}{12}t^3\right) dt = \left[3t - t^2 + \frac{1}{2}t^3 - \frac{23}{48}t^4\right]_{t=0}^{t=1} = 3 - 1 + \frac{1}{2} - \frac{23}{48} = \frac{97}{48}$.

Part (d): The alternating series error bound is the absolute value of the first omitted term of the series for $h(1)$. $\int_0^1 \left(\frac{54}{4!}t^4\right) dt = \left[\frac{9}{20}t^5\right]_{t=0}^{t=1} = \frac{9}{20}$. Error $\leq \left|\frac{9}{20}\right| = 0.45$.

Scoring Guide

Part (a): 2 points: {1 point: two terms; 1 point: remaining terms} Part (b): 2 points: {1 point: three terms for e^x ; 1 point: three terms for $e^x f(x)$ } Part (c): 2 points: {1 point: antiderivative; 1 point: answer} Part (d): 3 points: {1 point: uses fourth-degree term of Maclaurin series for f ; 1 point: uses first omitted term of series for $h(1)$; 1 point: error bound}